Chapter 14, Problem 10P

Problem

Design a problem to help other students better understand how to determine the Bode magnitude and phase plots of a given transfer function in terms of $j\omega$.

Step-by-step solution

Step 1 of 9

Bode plot is defined only for minimum phase systems.

That is, all poles and zeroes of the circuit must lie on the left side of the $j\omega$ axis only.

$$H(j\omega)=\frac{200}{(j\omega+4)(j\omega+5)}$$

Convert the transfer function to standard form.

$$H(j\omega)=\frac{10(j\omega)}{\left(\frac{j\omega}{4}+1\right)\left(\frac{j\omega}{5}+1\right)}$$

Step 2 of 9

There are two simple poles and a zero at the origin.

The corner frequencies are, $\omega=4,5$.

Every simple pole contributes -20dB slope to the Bode plot.

Every zero of first order contributes $+20\text{dB}$ slope to the Bode plot.

Step 3 of 9

Follow the steps to draw a magnitude plot.

1. Determine the magnitude and phase part of the transfer function.
2. First determine the gain contribution by the constant, $20\log K$.
3. Identify the poles or zeroes at the origin; they alter the gain contributed by the constant.
4. Calculate the change in gain due to the poles or zeroes at origin.
5. Identify the corner frequencies.
6. If the corner frequency belongs to a pole, add a slope of -20dB to the plot; if it belongs to a zero add a slope of $+20\text{dB}$ to the plot.

Step 4 of 9

The magnitude part of the transfer function is,

$$H_{\text{dB}} = 20\log 10 + 20\log|j\omega| - 20\log\left|1+\frac{j\omega}{4}\right| - 20\log\left|1+\frac{j\omega}{5}\right|$$

The phase part of the transfer function is,

$$\phi = 90° - \tan^{-1}\left(\frac{\omega}{4}\right) - \tan^{-1}\left(\frac{\omega}{5}\right)$$

Step 5 of 9

The contribution by the constant 10 is,

$$20\log 10 = 20\text{dB} \quad (\log_{10} 10 = 1)$$

The contribution to the plot by a zero at the origin is,(At $\omega = 1$ the contribution is zero, since the origin and $\omega = 1$ are assumed to be a decade apart) $-20dB$ at zero corner frequency.

Thus, the magnitude ta zero frequency is 0dB.

Step 6 of 9

Since there is a zero at the origin, the initial slope of the plot is $+20\text{dB}$.

At the corner frequency $\omega = 4$, a pole is present.

This contributes -20dB slope to the plot.

Since the initial slope is $+20\text{dB}$ an addition of -20dB makes the slope 0dB.

At the corner frequency $\omega = 5$, a pole is present.

Since the initial slope is 0dB an addition of -20dB makes the slope -20dB.

Step 7 of 9

Draw the magnitude plot.

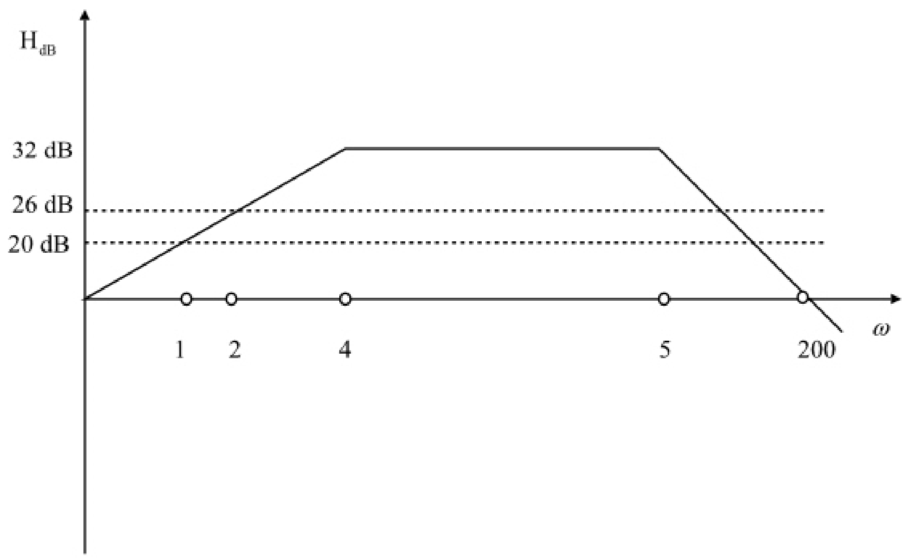

Figure 1: Magnitude plot

Step 8 of 9

The expression for the phase of the transfer function is,

$$\phi = 90° - \tan^{-1}\left(\frac{\omega}{4}\right) - \tan^{-1}\left(\frac{\omega}{5}\right)$$

At $\omega = 0$ the phase of the system is $90°$.

At $\omega = 1$ the phase of the system is $64.6°$.

At $\omega = 2$ the phase of the system is $41.63°$.

At $\omega = 4$ the phase of the system is $6.34°$.

At $\omega = 5$ the phase of the system is $-6.34°$.

At $\omega = 200$ the phase of the system is $-87.42°$.

Step 9 of 9

Draw the Bode plot, both magnitude and phase plot together.

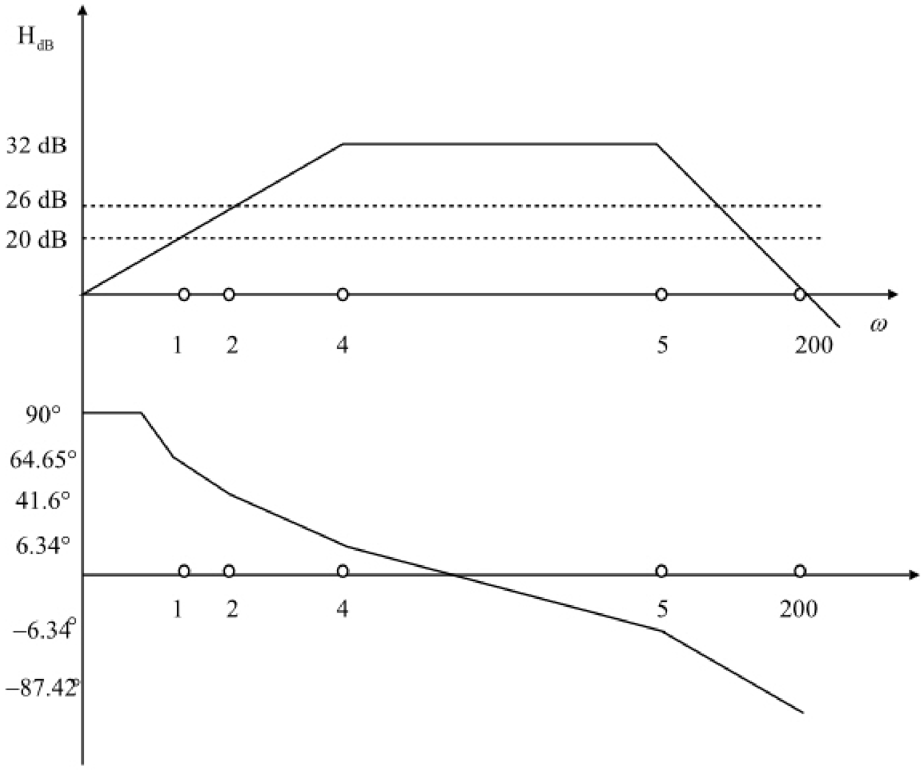

Figure 2: Bode plot

Thus, the bode plot of the system is drawn.